

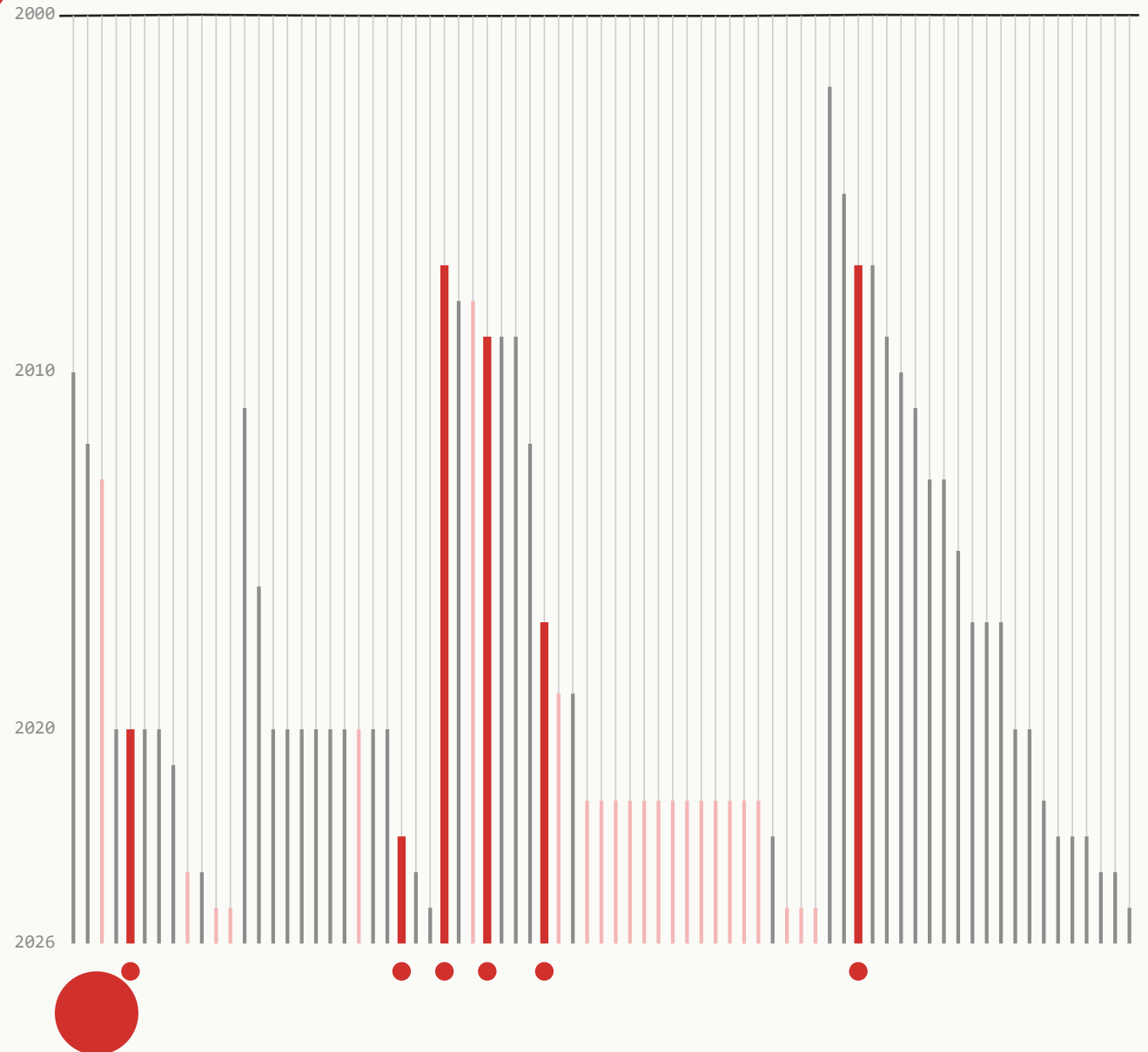
Who owns the flood?

A sketch deck of the interaction states and the first prototype of our data visualization.

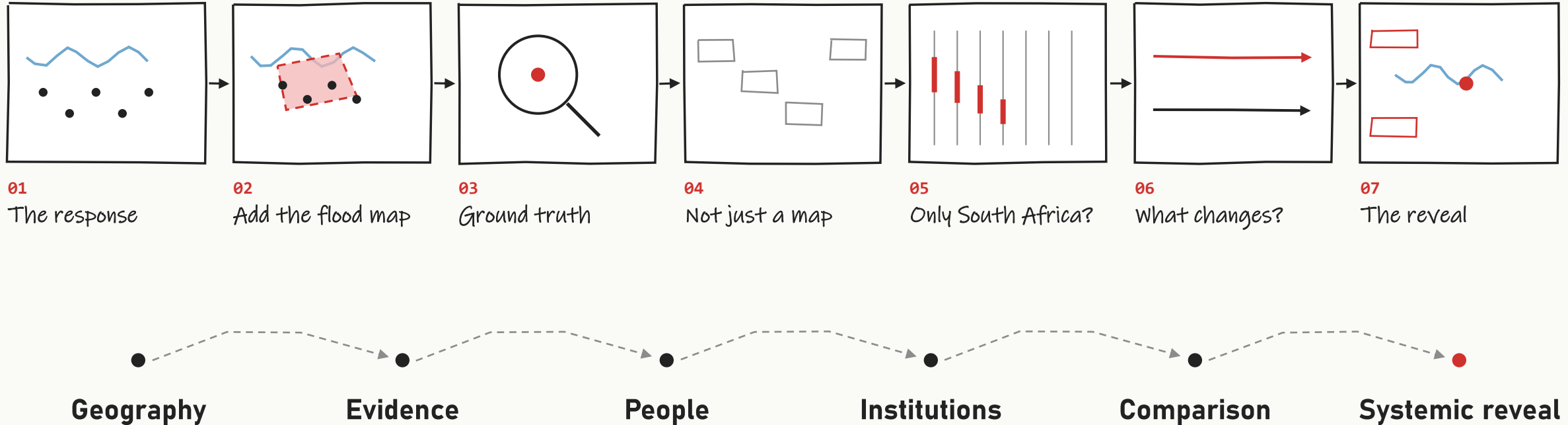
Hollie & Wandile · Harvard GSD MDE · Sept 2026

Each vertical line is one of South Africa's 75 dated "temporary" relocation sites. It runs from the year the site opened to 2026. Pink = opened for flood survivors. Red = flooded again after people moved in (7 sites; 30 of 84 searched).

Source: SA TRAs 2022-2026 site list; reflooding sweep (data/tra_sites_exposure.csv)

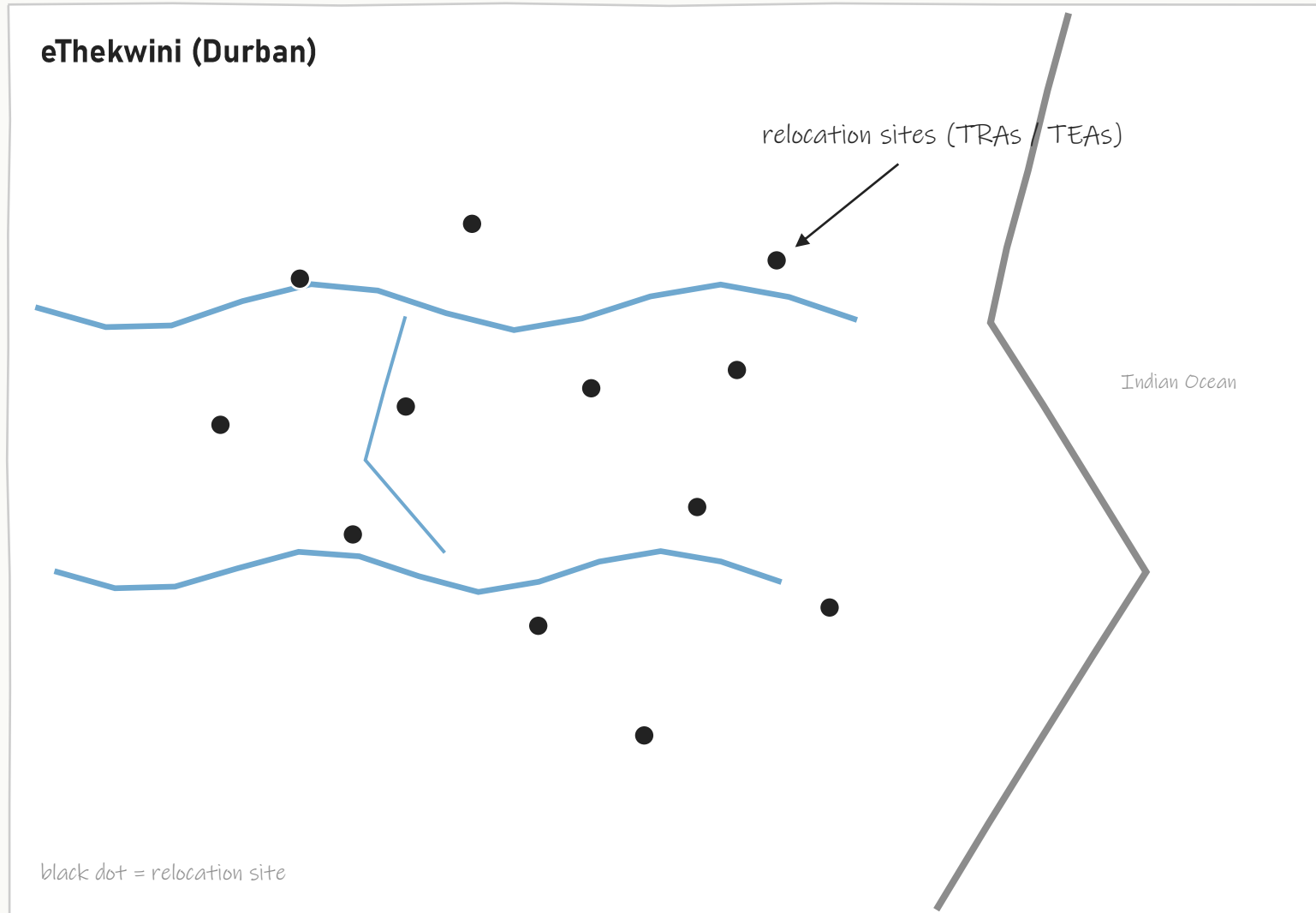


Seven states, one scroll



The viewer scrolls. Each state adds one layer to the last: the map, then the evidence, then the people, then the institutions behind it, then the world, and finally the reveal.

Frame 1 — The response



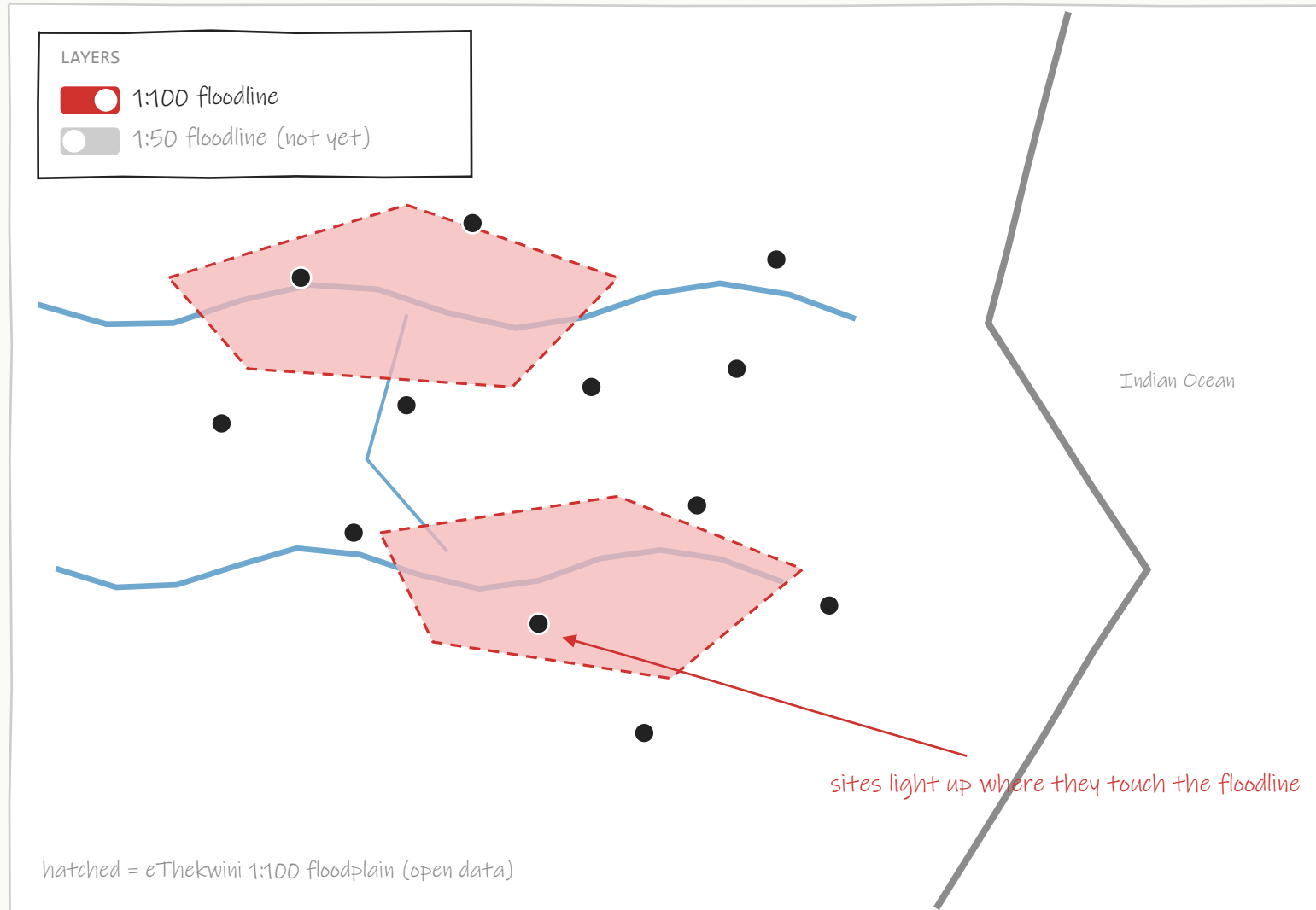
01

After the 2022 floods, government identified and relocated displaced residents.

INTERACTION

The map loads. Relocation sites fade in one by one as the viewer scrolls.

Frame 2 — Add the flood map



02

Same map. Add the city's 1:100 floodline (1:50 not yet obtained).

INTERACTION

Toggle the floodline layer. Relocation sites become visible against known flood risk.

Are these places actually safer?

Frame 3 — Ground truth

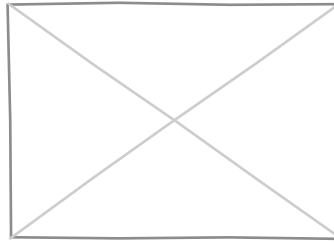
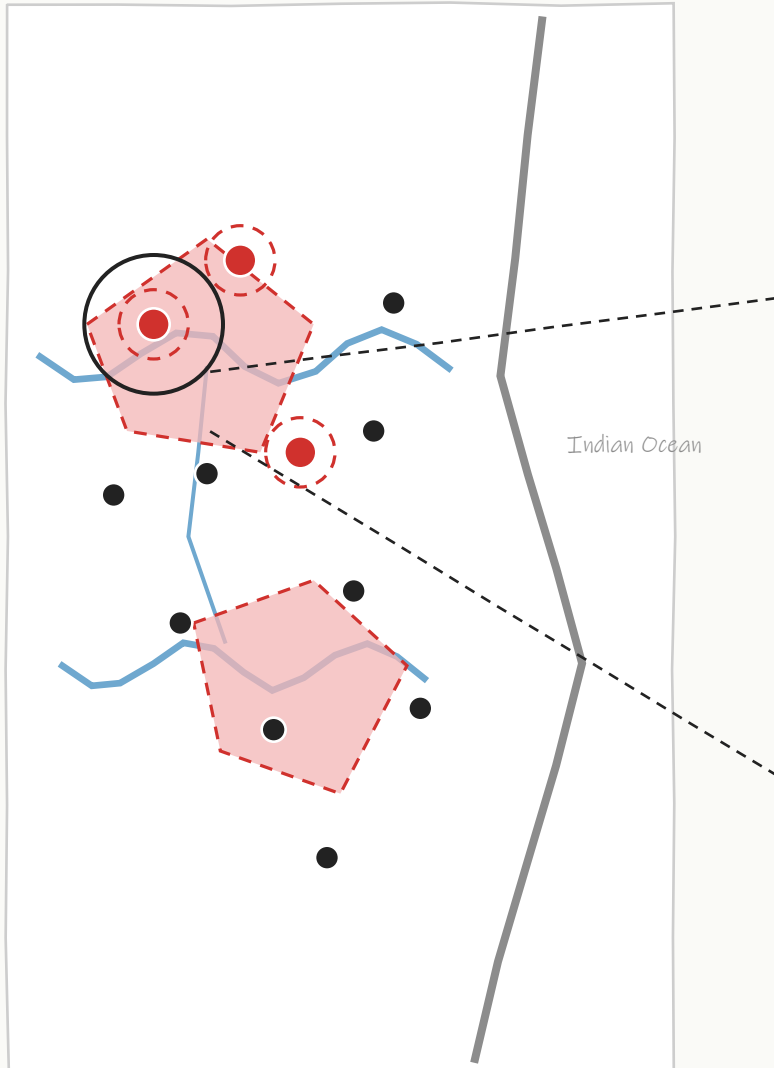


photo: Gwala St, 2022

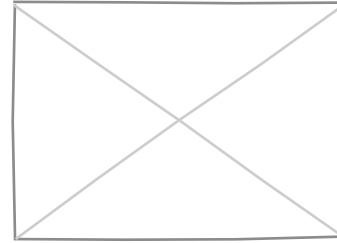


photo: Isipingo

"We have been moved from one place to another."

resident, GroundUp (headline)

INCIDENTS AT THIS CAMP



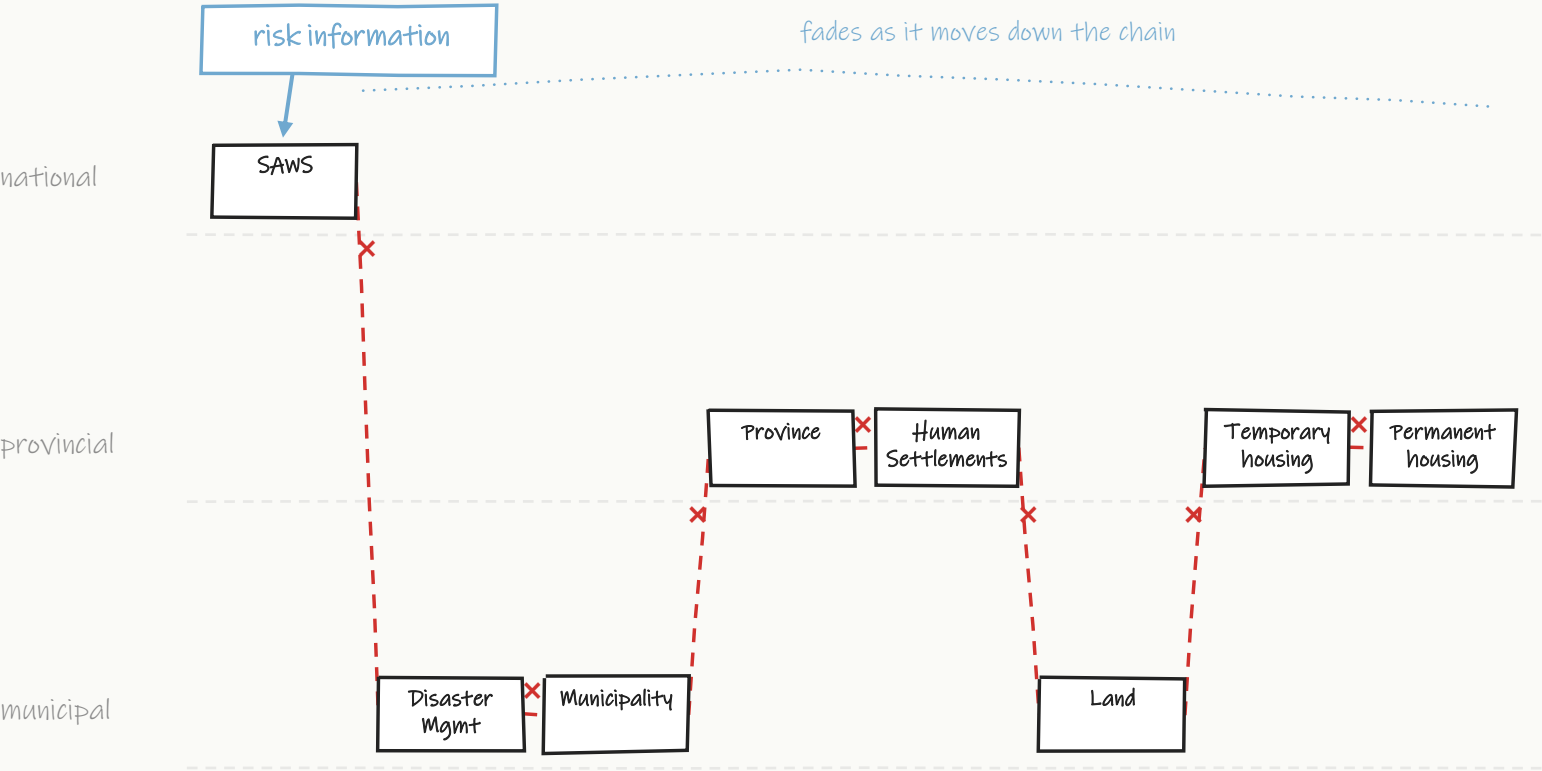
03

Some relocated communities experienced flooding again. Gwala St (Lamontville), Isipingo and Madamfana are documented Durban examples; nationally, 7 sites (30 of 84 searched).

INTERACTION

Zoom into a site: photographs, testimony and the flood incidents open in a side panel.

Frame 4 — This is not just a map problem



04

The risk information exists, but it passes through a fragmented decision chain.

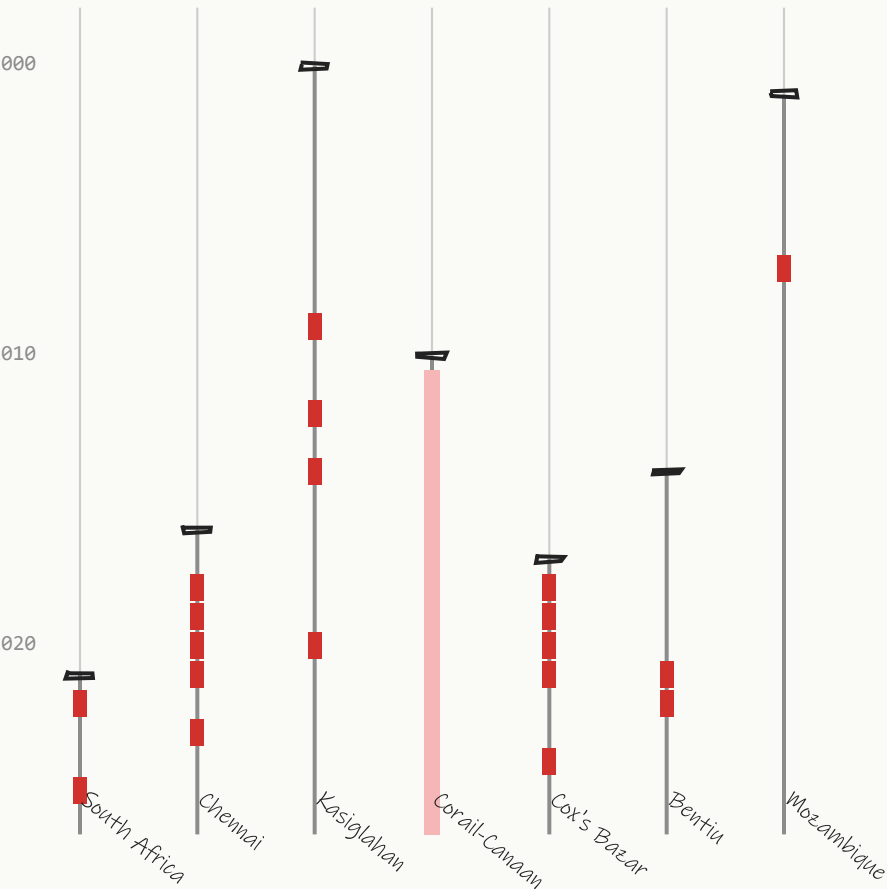
INTERACTION

Pull back from the sites. The chain draws itself link by link; a red x appears at every hand-off.

8 links • 3 spheres • 7 hand-offs • 0 end-to-end owners

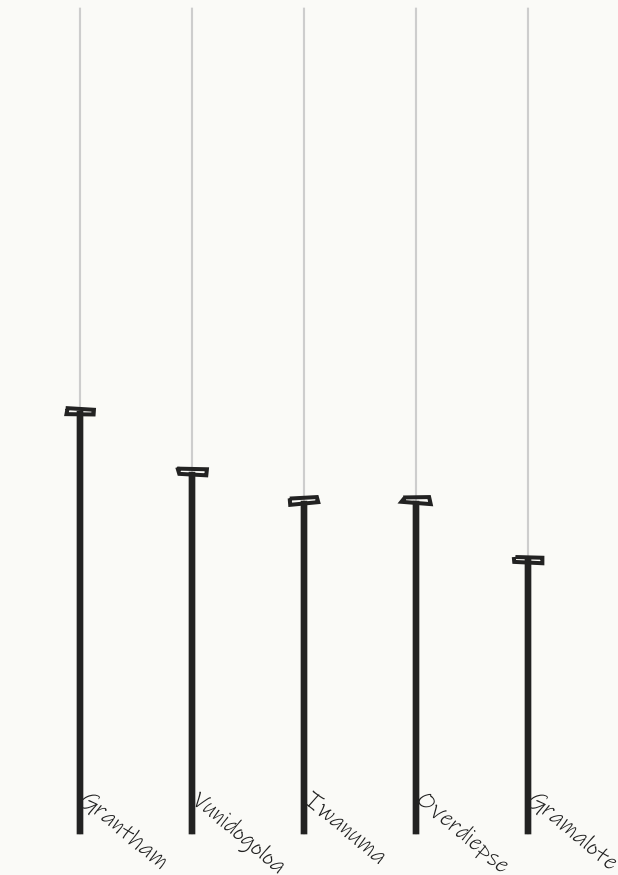
Frame 5 — Is this only South Africa?

Failure-like systems



Different architectures

dash = move-in; red = flood after the move



05

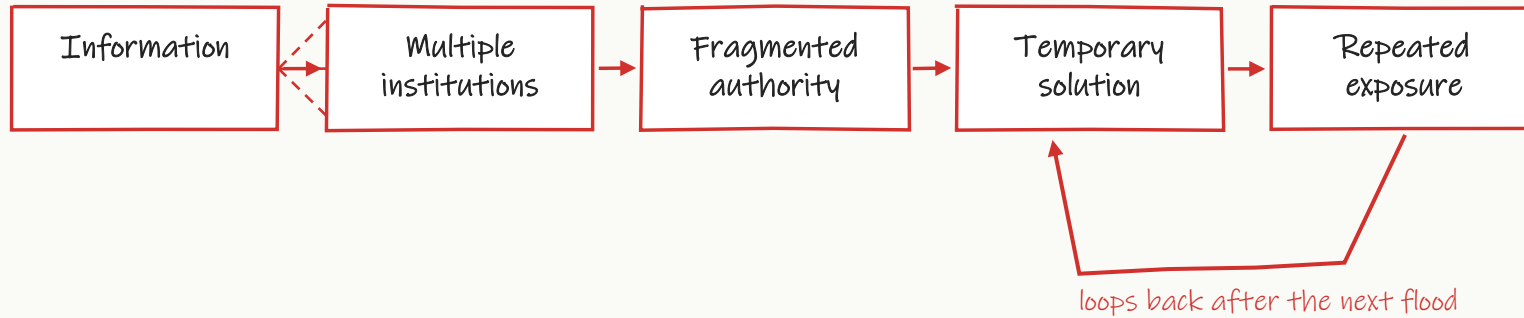
International cases: failure-like systems on the left, different architectures on the right.

INTERACTION

Scroll: red flood marks drop onto each timeline after the move. The right-hand timelines stay clean.

Frame 6 — What changes?

Failure pattern



Alternative pattern



Separates the cases: one owner (0% vs 90%), residents choose the site (0% vs 90%), binding flood maps (8% vs 100%).

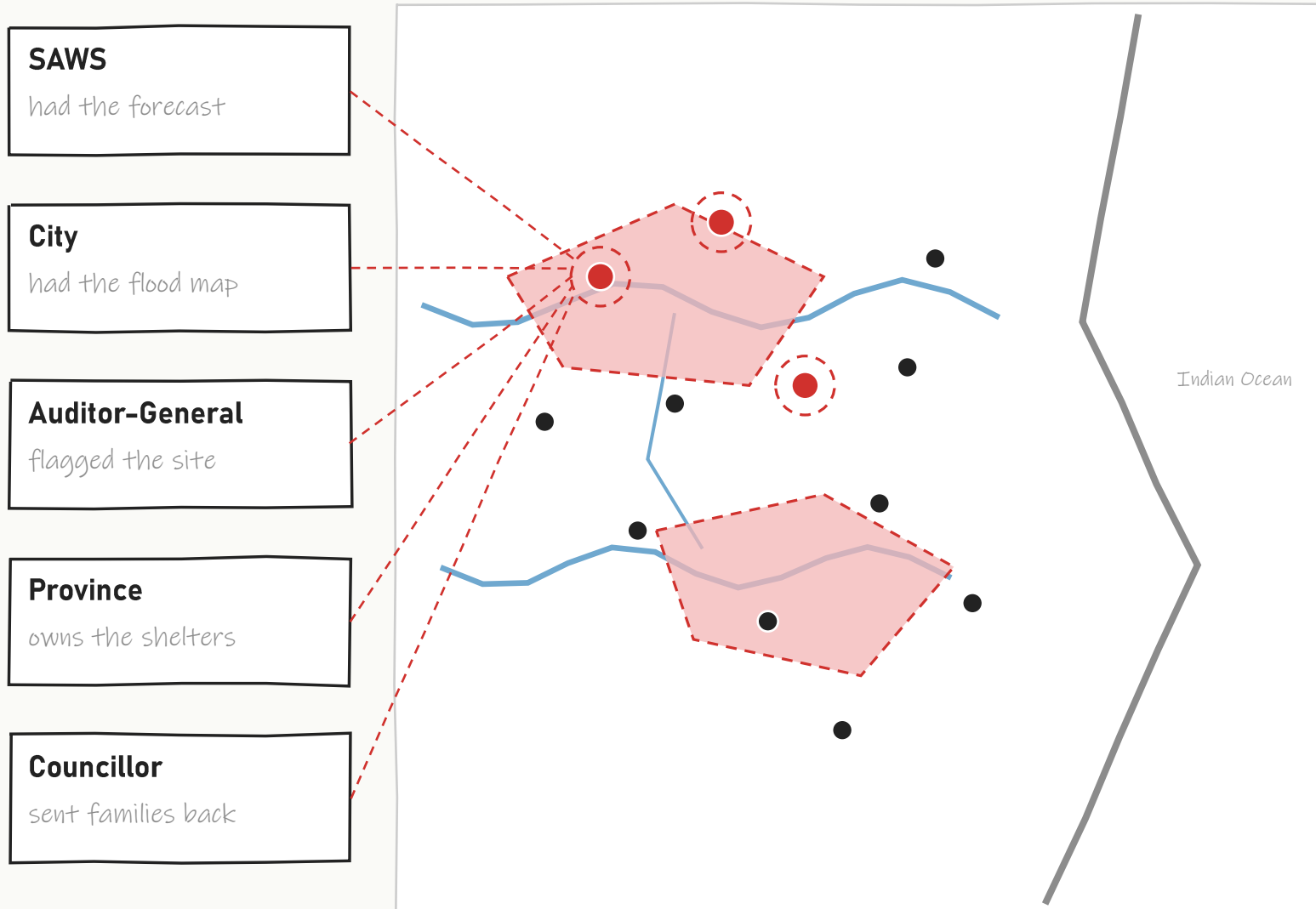
06

Compare the two decision architectures.

INTERACTION

The two flows animate in parallel. The failure flow loops back on itself.

Frame 7 — The reveal



07

Return to the original Durban map, with the system drawn over it.

INTERACTION

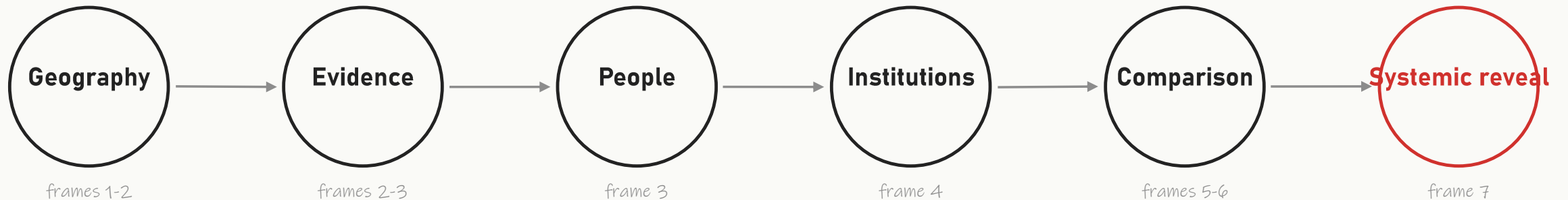
Each institution lights up and links to what it knew. ~~None connects to the decision about the site.~~
~~Why are people living in dangerous places?~~

Who owns the decision that determines where they live?

South Africa has the information. The institutions that possess it do not control the decisions that determine exposure.

What are we actually building?

An interactive scroll story that moves between:

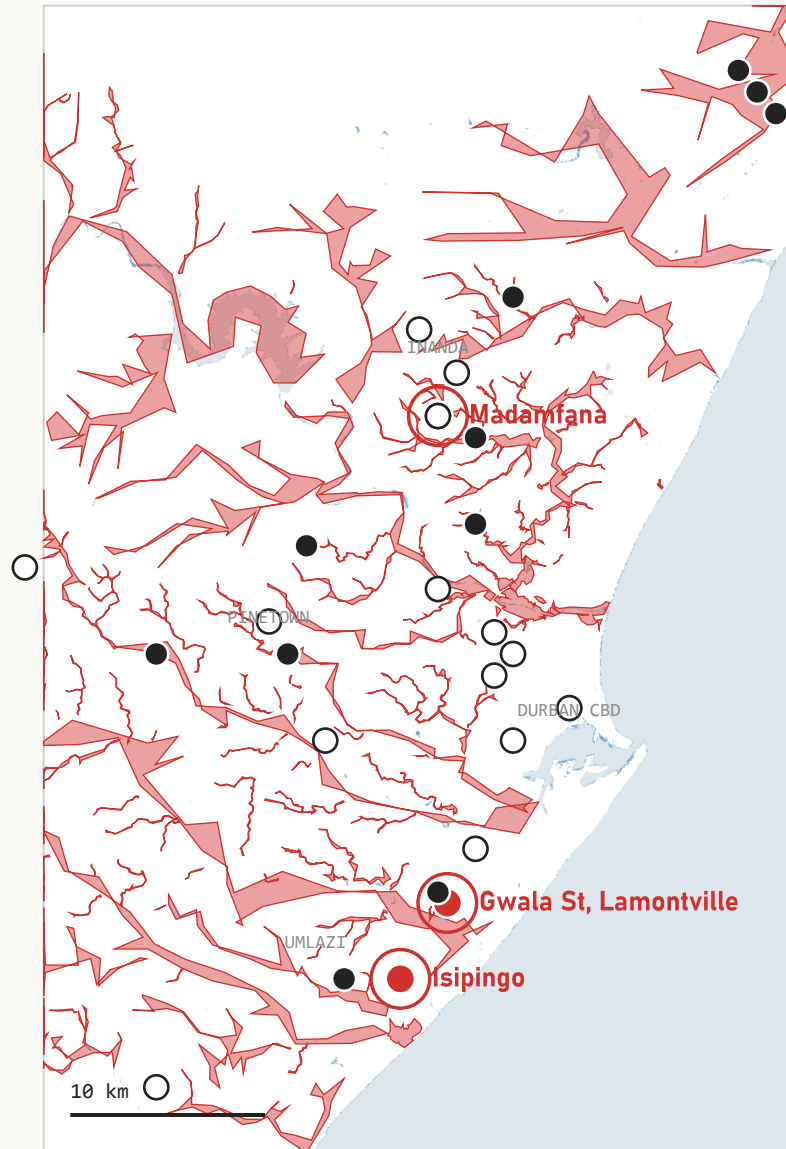


HIGH-FIDELITY FIRST PROTOTYPE

Frame 2: the eThekwini map with relocation sites and floodline data.

Why this frame first: it uses our actual dataset (84 sites, the city floodplain, 300 random comparison points) and it establishes the visual language of the whole project: map, points, layers.

Are these places actually safer?



LAYERS (toggle)

- ☒ Satellite water, 1984-2021
- ☒ 1:100 floodline
- ☒ Relocation sites
- ☒ Documented reflooding
- ☐ 1:50 floodline (not yet)

11 of 28

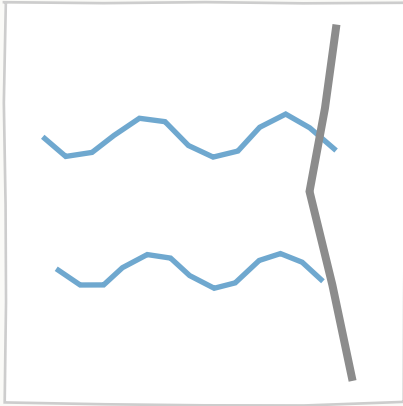
Durban relocation sites inside the city's 1:100 floodplain or within 250 m of it.

Random ground across Durban: 34%.

The flood map made no difference to where people were put (binomial $p = 0.34$).

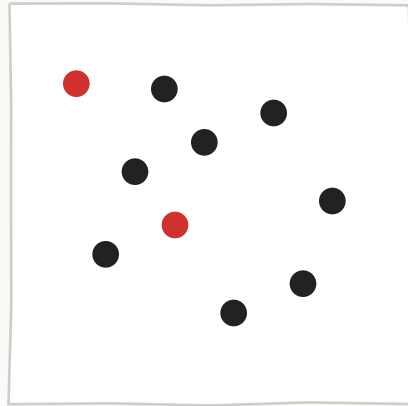
- site inside / within 250 m of floodplain
- site farther away
- ⊙ flooding documented after the move

Core visual typology



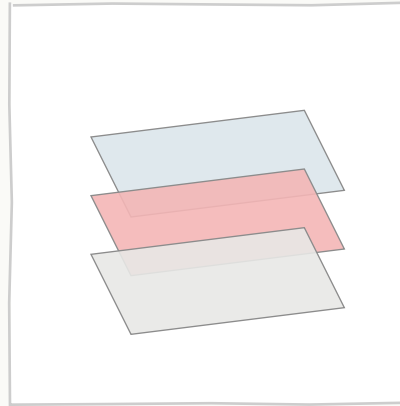
Map

= geography / exposure



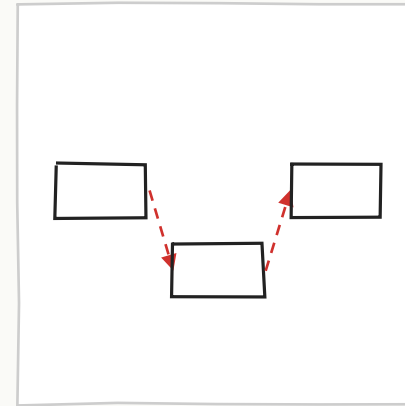
Points

= people / sites



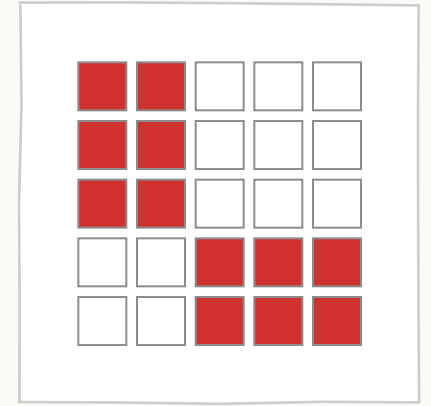
Layers

= evidence



Flow diagram

= institutional
responsibility



Comparison matrix

= system architecture

One visual language across the scroll: geography and evidence are layered on the map, then the story pulls back to flows (who is responsible) and a matrix (how the systems compare).